TCET
DEPARTMENT OF VOCATIONAL STUDIES
Choice Based Credit Grading Scheme with Holistic and Multidisciplinary Education
Under Autonomy - CBCGS-HME 2025
University of Mumbai

NAME: SAGARSUMAN P MANJHI **PRACTICAL: 06** **BATCH : 04**
SUBJECT: PYTHON PROGRAMMING LAB **SYBVOC :SEM-III (DS)**

Experiment Based on Functions and Modules

AIM : To study and implement user-defined functions, function arguments, return values, recursion, and built-in modules by developing a Restaurant Billing System that calculates the bill, applies GST and discount using functions, and generates a unique bill number using the random module.

LEARNING OBJECTIVE :

- To understand user-defined functions in Python.
- To learn function arguments and return values.
- To perform calculations using functions.
- To understand the use of built-in modules.
- To generate a random bill number using the random module.
- To develop a restaurant billing system using Python.

TOOLS :

Sr. No.	Name Of Resources	Specification	Quantity	Remarks
1.	Hardware	Computer (i5-i7) Ram (Min 16GB)	1	For All Practical
2.	Software	Python / IDLE / VS Code	1	For All Practical

THEORY :

1. Functions: A function is a reusable block of code that performs a specific task. It is defined using the def keyword.

Example:

```
def greet():    print("Welcome to Restaurant")
```

2. Function Arguments: Arguments are values passed to a function when it is called. They allow a function to work with different inputs.

Example:

```
def calculate_total(price, quantity):    return price * quantity
```

3. Return Statement: The return statement sends a result back to the calling function.

Example:

```
def square(n):    return n * n
```

4. Recursion: Recursion is a technique in which a function calls itself to solve a problem. A base condition is required to stop recursive calls.

Example:

```
def factorial(n):    if n <= 1:        return 1    return n * factorial(n - 1)
```

5. Modules: A module is a Python file containing functions and other code that can be reused in a program. Python provides built-in modules such as random and math.

Example:

```
import randomprint(random.randint(1000, 9999))
```

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Source Code:

```python
# PRACTICAL 6
import random

def calculate_total(price, quantity):
    return price * quantity
def calculate_discount(amount):
    if amount >= 1000:
        return amount * 0.10
    return 0
def calculate_gst(amount):
    return amount * 0.05

def print_bill(customer, item, subtotal, discount, gst, total, bill_no):
    print("\n--- RESTAURANT BILL ---")
    print("Bill Number:", bill_no)
    print("Customer Name:", customer)
    print("Item:", item)
    print("Subtotal: Rs.", round(subtotal, 2))
    print("Discount: Rs.", round(discount, 2))
    print("GST: Rs.", round(gst, 2))
    print("Final Total: Rs.", round(total, 2))

customer = input("Enter customer name: ")
item = input("Enter food item: ")
price = float(input("Enter price: Rs. "))
quantity = int(input("Enter quantity: "))

if price >= 0 and quantity > 0:
    subtotal = calculate_total(price, quantity)
    discount = calculate_discount(subtotal)
    gst = calculate_gst(subtotal - discount)
    total = subtotal - discount + gst
    bill_no = random.randint(1000, 9999)

    print_bill(customer, item, subtotal, discount, gst, total, bill_no)
else:
    print("Invalid price or quantity")
```

Output:

```
PS C:\Users\sagar\Documents\PYTHON> python Practicals/P6.py
Enter customer name: Sagarsuman Manjhi
Enter food item: Shahi Paneer
Enter price: Rs. 299
Enter quantity: 4

--- RESTAURANT BILL ---
Bill Number: 7900
Customer Name: Sagarsuman Manjhi
Item: Shahi Paneer
Subtotal: Rs. 1196.0
Discount: Rs. 119.6
GST: Rs. 53.82
Final Total: Rs. 1130.22
PS C:\Users\sagar\Documents\PYTHON>
```

Result and Discussion:

The Restaurant Billing System was implemented successfully using Python functions and the random module.The program accepts customer details, calculates the subtotal, applies a 10% discount when the subtotal is ₹1,000 or more, calculates 5% GST on the discounted amount, generates a random four-digit bill number, and displays the final bill.

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Learning Outcomes:

- Learned to define and call user-defined functions.
- Understood function arguments and return values.
- Used the random module to generate a bill number.
- Practised input validation and arithmetic operations.
- Developed a modular Python application for restaurant billing.

Course Outcomes:

- Understand Python programming fundamentals and functions.
- Apply function arguments and return values to solve problems.
- Use built-in modules to perform specific operations.
- Develop a modular Python application for a real-world problem.

CONCLUSION :

The Restaurant Billing System was developed successfully using Python functions and the built-in random module. The program calculates the subtotal, applies the eligible discount, calculates GST, generates a random bill number, and displays the final amount. This practical demonstrates how modular programming can be used to solve a real-world billing problem.

VIVA QUESTIONS :

Q1. What is a function in Python?
A function is a reusable block of code designed to perform a specific task.

Q2. Which keyword is used to define a function?
The def keyword is used to define a function.

Q3. What are function arguments?
Arguments are values passed to a function when it is called.

Q4. What is the purpose of the return statement?
It returns a value from a function to the calling code.

Q5. What is a Python module?
A module is a file containing Python code, such as functions and variables, that can be imported and reused.

FOR FACULTY USE ONLY :

Correction Parameters	Formative Assessment [40%]	Timely completion of practical [40%]	Attendance/ Learning Attitude [20%]	
Marks Obtained				